

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1895

Loving South 138 kV
October 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1895. The Interconnection Customer proposes a 1232.9 MW natural gas combined cycle facility interconnecting at the Loving South 138 kV substation in the Virginia area. The request was received on October 04, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 451 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$59,868,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Borden to Barneys 138.0 kV	Lovng South to Custer 138.0 kV	527.0	215	100.0
Borden Substation to Loving S 138.0 kV	Loving S to Custer 138.0 kV	433.3	271	100.0
Custer to Loving South 138.0 kV	Brden to Loving S 138.0 kV	409.3	287	100.0
Occoquan to Cuter Substation 138.0 kV	Custer to Jeune Landing 138.0 kV Line	319.1	232	65.7
Custer to Jeun Landing 138.0 kV	Custer to Occoquan 138.0 kV	318.4	235	65.1
Idaho Sprigs to Barneys 138.0 kV	Custer Substation to Loving South 138.0 kV L	303.4	215	64.3
Sideburn to Keene Mill 138.0 kV	Custer to Loving S 138.0 kV	242.1	177	35.7
Port Saint Joe to Keene Mill 138.0 kV	Loving South to Custer 138.0 kV	231.6	197	35.7
Covanta Fairfax Energy to Fairfax 138.0 kV Line	Jeune Landing to Fairfax 138.0 kV	202.5	175	20.1
Fairfax Substation to Franconia 138.0 kV	Loving S to Borden 138.0 kV	200.4	289	38.3
Barneys to Port St Joe 138.0 kV	Custer to Loving South 138.0 kV	196.2	226	35.7
Jeune Landing to Fairfax 138.0 kV	Covanta Fairfax Energy to Occoquan 138.0 kV	182.3	298	28.1
Brookride to Franconia 138.0 kV line	Borden to Loving S Substation 138.0 kV	181.5	304	38.3
Occoquan Substation to Covanta Fairfax Energy 138.0 kV line	Jeune Landing to Fairfax 138.0 kV	156.6	273	20.1
Agleton to Idaho Springs 138.0 kV	Loving S to Custer 138.0 kV	155.4	232	29.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
S Houston to Peavy Fall Hy 138.0 kV Line	Loving S to Custer 138.0 kV	142.1	310	28.6
Brookridge to Burrows Crossing Road 138.0 kV	Nestuca to Carreon 138.0 kV	141.8	376	18.6
Nestuca to Carreon 138.0 kV line	Burrows Crossing Rd to Brookridge 138.0 kV	139.6	259	12.2
Luna to Peavy Fall Hy 138.0 kV	Custer to Loving S 138.0 kV line	135.6	367	28.6
Angleton to Schoolcraft 138.0 kV Line	Loving S to Custer 138.0 kV Line	132.6	242	29.1
Luna to Yavapai 138.0 kV Line	Custer to Loving South 138.0 kV L	131.9	411	28.6
Jeune Landing to Occoquan 138.0 kV	Fairfax to Jeune Landing 138.0 kV	131.7	221	25.9
Falu 1334 to Burnt Church 138.0 kV	Van Dom to Hayfield 138.0 kV line	131.1	272	5.3
Burnt Church to Donaldson Creek 138.0 kV	Van Dom to Hayfield 138.0 kV	125.4	231	5.3
Sideburn to Yavapai 138.0 kV	Custer to Loving South 138.0 kV	119.2	380	23.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Borden to Barneys 138.0 kV	Rebuild line with higher-capacity conductor	\$1,455,000	100% (\$1,455,000)
Borden Substation to Loving S 138.0 kV	Rebuild line with higher-capacity conductor	\$5,195,000	100% (\$5,195,000)
Custer to Loving South 138.0 kV	Rebuild line with higher-capacity conductor	\$1,039,000	100% (\$1,039,000)
Occoquan to Custer Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$3,037,000	95% (\$2,891,000)
Custer to Jeun Landing 138.0 kV	Rebuild line with higher-capacity conductor	\$1,637,000	96% (\$1,572,000)
Idaho Sprigs to Barneys 138.0 kV	Rebuild line with higher-capacity conductor	\$2,309,000	95% (\$2,200,000)
Sideburn to Keene Mill 138.0 kV	Rebuild line with higher-capacity conductor	\$2,831,000	90% (\$2,552,000)
Keene Mill to Port Saint Joe 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,448,000	90% (\$1,306,000)
Covanta Fairfax Energy to Fairfax 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,210,000	89% (\$2,856,000)
Fairfax Substation to Franconia 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,581,000	94% (\$1,494,000)
Port Saint Joe to Barneys 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,062,000	92% (\$2,811,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jeune Landing Substation to Fairfax 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,762,000	90% (\$1,586,000)
Brookride to Franconia 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,506,000	94% (\$1,424,000)
Occoquan to Covanta Fairfax Energy 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,560,000	89% (\$4,946,000)
Agleton to Idaho Springs 138.0 kV	Rebuild line with higher-capacity conductor	\$3,902,000	81% (\$3,149,000)
S Houston to Peavy Fall Hy 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,109,000	81% (\$894,000)
Brookridge to Burrows Crossing Road 138.0 kV	Rebuild line with higher-capacity conductor	\$1,662,000	89% (\$1,482,000)
Nestuca to Carreon 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,397,000	82% (\$1,960,000)
Luna to Peavy Fall Hy 138.0 kV	Rebuild line with higher-capacity conductor	\$1,849,000	81% (\$1,490,000)
Angleton to Schoolcraft 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,290,000	81% (\$4,269,000)
Luna to Yavapi 138.0 kV	Rebuild line with higher-capacity conductor	\$1,737,000	81% (\$1,399,000)
Occoquan to Jeune Landing 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,510,000	98% (\$2,447,000)
Falu 1334 to Burnt Church 138.0 kV	Rebuild line with higher-capacity conductor	\$6,065,000	100% (\$6,065,000)
Burnt Church to Donaldson Creek 138.0 kV	Rebuild line with higher-capacity conductor	\$1,859,000	100% (\$1,859,000)
Sideburn to Yavapai 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,811,000	84% (\$1,527,000)
	Total allocated to project		\$59,868,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.